

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S5 (PT) (R, S) Examination December 2023 (2019 Scheme)

Course Code: EET435**Course Name: RENEWABLE ENERGY SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | Differentiate between Conventional and Non-conventional sources of energy | (3) |
| 2 | Explain 3 applications of stand-alone solar PV systems | (3) |
| 3 | Differentiate between Open Cycle and Closed Cycle OTEC systems | (3) |
| 4 | Explain the environmental impacts of Tidal Energy conversion systems | (3) |
| 5 | List 3 important criteria for the selection of site for a WECS | (3) |
| 6 | A wind turbine rotor has a blade diameter of 20m. At a wind speed of 16m/s the turbine produces a power output of 270kW. Given that the density of air is 1.2 kg/m ³ determine the maximum power coefficient of the turbine. | (3) |
| 7 | What is the product obtained from Biomass gasification? Name the stages involved in the gasification process | (3) |
| 8 | Name the process used to produce biogas from wet biomass? Explain the stages involved in this process. | (3) |
| 9 | List three methods for storing hydrogen | (3) |
| 10 | What are the benefits of using hydrogen as a fuel? | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Draw a fully labelled sketch and explain the working of a Solar Thermal Electric Power generation system using Central Tower solar-collector. | (10) |
| | b) Draw a fully labelled I-V Curve of a solar cell indicating all relevant voltages and currents. Also indicate the voltage and current corresponding to the maximum power point and show how the Fill Factor can be computed from this curve. | (4) |
| OR | | |
| 12 | a) With a neat labelled diagram explain the construction and working of a Flat plate solar collector | (10) |

- b) Compare the features (4 points) of non-concentrating and non-concentrating type solar collectors. (4)

Module II

- 13 a) With a neat diagram explain the working of a Hybrid Cycle OTEC system (10)
b) Explain the advantages of OTEC systems (4)

OR

- 14 a) Draw a schematic diagram and explain the working of a double-basin Tidal power Plant (10)
b) What are the environmental impacts of Tidal Power plants? (4)

Module III

- 15 a) Explain in detail the different types of classifications of Wind energy Conversion systems (10)
b) Explain how wind speed is measured using an anemometer (4)

OR

- 16 a) Determine the theoretical maximum power that can be extracted from wind using a wind turbine. Also show that this condition is achieved when the downstream velocity is one-third of the upstream velocity. (10)
b) Explain the factors to be considered for selection of site for WECS (4)

Module IV

- 17 a) How can Urban waste be converted to useful energy? Draw a neat diagram and explain the step-by-step process (10)
b) Explain the process of converting Biomass to Ethanol (4)

OR

- 18 a) What is KVIC model of Biogas generation? With a neat schematic diagram explain its working. (10)
b) Explain any four significant factors which affect the performance of a Biogas plant. (4)

Module V

- 19 a) Draw a neat schematic of a Hydrogen-Oxygen fuel cell, write down the chemical reactions involved and explain the working. (10)
b) Explain any 2 methods used for storage of Hydrogen (4)

OR

- 20 a) What are the different types of turbines used in Small Hydro power plants? With neat sketch explain any two such turbines (10)
- b) Explain how Hydrogen is produced by Electrolysis (4)
